

Unit 6, Lesson 5: Proportional Relationships and Similar Triangles

Benchmark

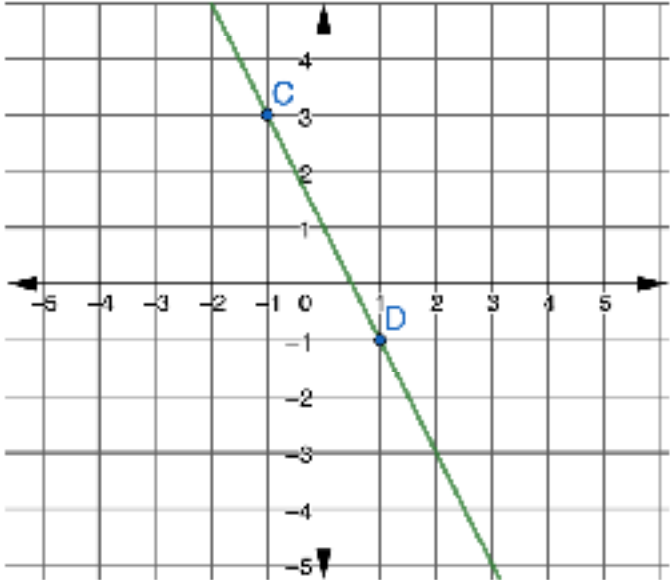
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Learning Target

☐ I can solve mathematical problems using similar triangles.

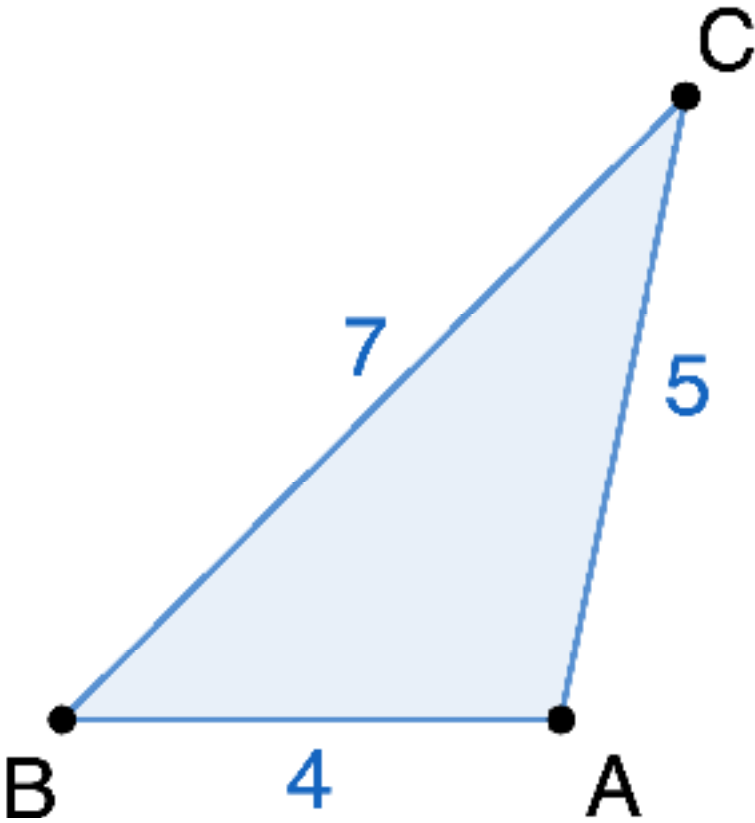
6.5.1 Warm-Up: Finding Slope

1. Find the slope of each line using the given information.

Line	Slope
	
A line that passes through (4, 1) and (10, 4).	

6.5.2 Exploration: Ratios & Similar Triangles

Triangle ABC is shown.



Lauren wants to create three new triangles that have the same shape as triangle ABC , but are different sizes. She multiplies each side of triangle ABC by the scale factors given in the table to determine the sides of the new triangles.

1. Complete the table to determine the side lengths of the three new triangles: triangle *DFH*, triangle *GKM*, and triangle *SLN*.

Triangle	Scale Factor	Length of Short Side	Length of Medium Side	Length of Long Side
<i>ABC</i>	1	4	5	7
<i>DFH</i>	2			
<i>GKM</i>	3			
<i>SLN</i>	$\frac{1}{2}$			

2. Complete the table by writing the following ratios.

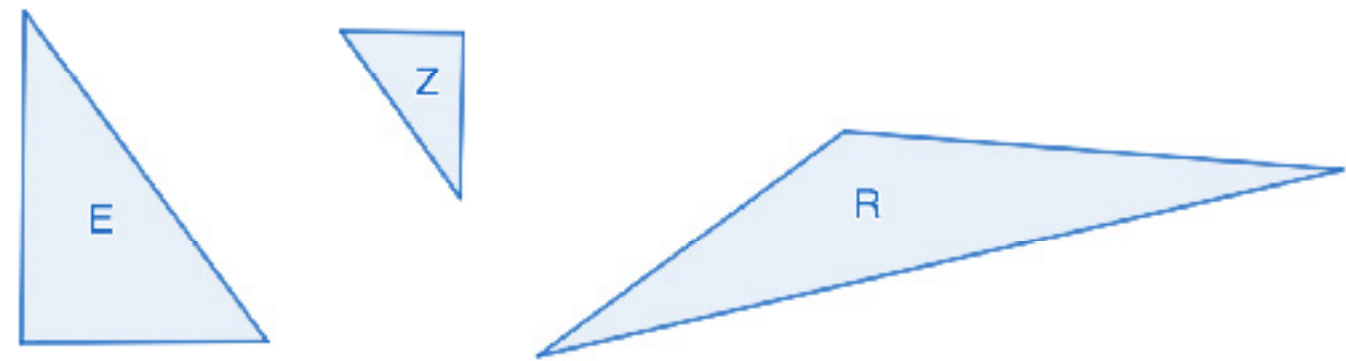
Triangle	$\frac{\text{long side}}{\text{short side}}$	$\frac{\text{long side}}{\text{medium side}}$	$\frac{\text{medium side}}{\text{short side}}$
<i>ABC</i>	$\frac{7}{4}$ or 1.75		
<i>DFH</i>			
<i>GKM</i>			
<i>SLN</i>			

3. Compare your answers with your partner. Then, discuss your observations of the ratios.

6.5.3 Guided Instruction: Similar Triangles and Proportionality

In the Exploration, the triangles created are called **similar triangles**. In mathematics, similar figures have the same shape, but are not necessarily the same size.

1. Three triangles are shown.

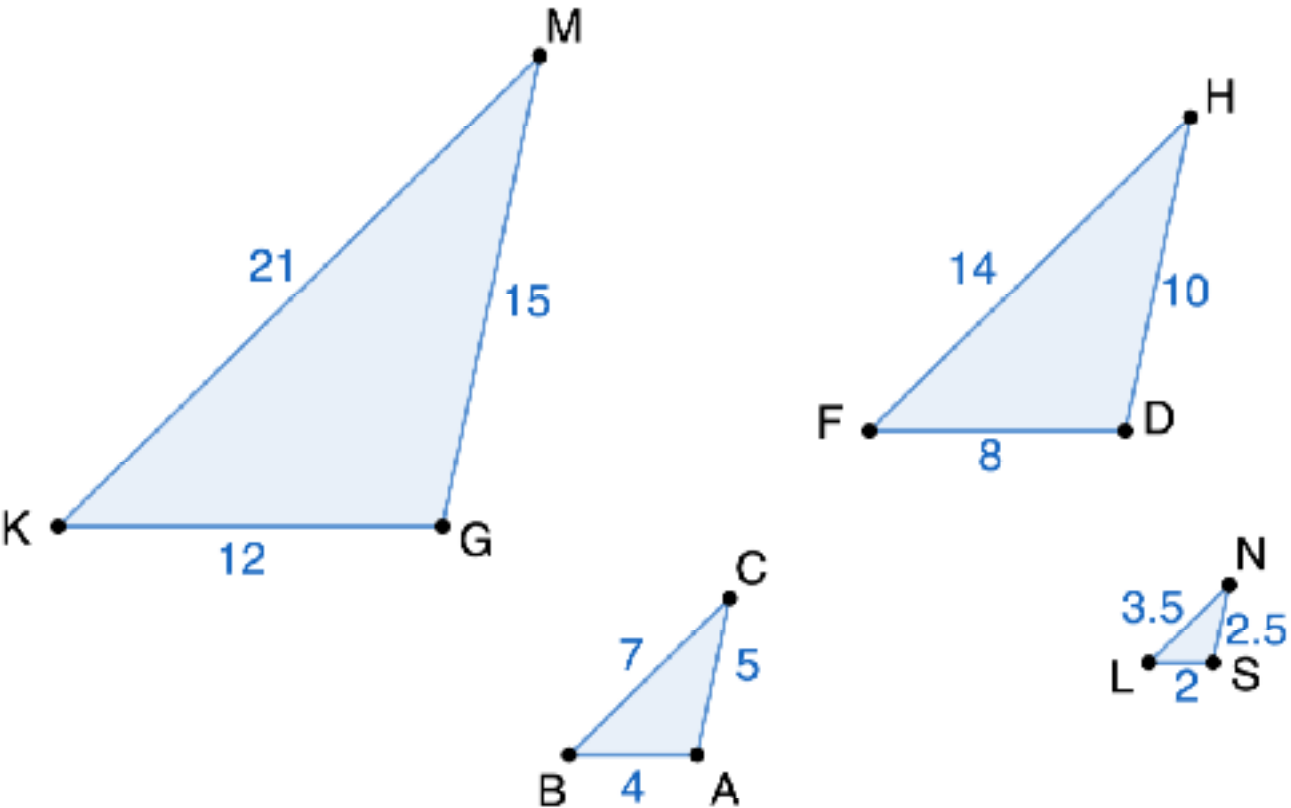


Discuss with your partner which two triangles appear to be similar.

Similarity cannot be determined by appearance alone. Triangles are similar if their corresponding sides are proportional.

The Exploration used the words “long,” “medium,” and “short” to describe the sides of the triangles. Sides of similar triangles that are paired are called **corresponding sides**.

Similar triangles ABC , DFH , GKM , and SLN from the Exploration are shown.

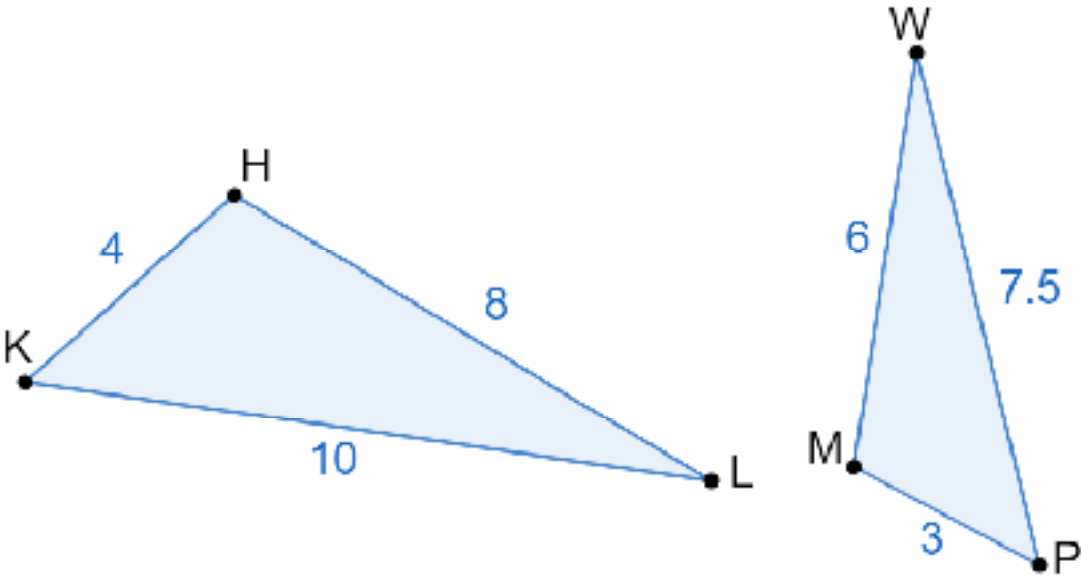


In the Exploration, the long side of triangle ABC is the side between points B and C , called segment BC .

2. Complete the table by naming the corresponding long sides of the triangles from the Exploration.

Triangle	Long Side
Triangle DFH	Segment _____
Triangle GKM	Segment _____
Triangle SLN	Segment _____

3. Triangle HKL and triangle MPW are similar.



a. Complete each statement to match the corresponding sides in the two similar triangles.

- Segment HK corresponds with segment _____.
- Segment HL corresponds with segment _____.
- Segment KL corresponds with segment _____.



In similar triangles, corresponding sides are proportional.

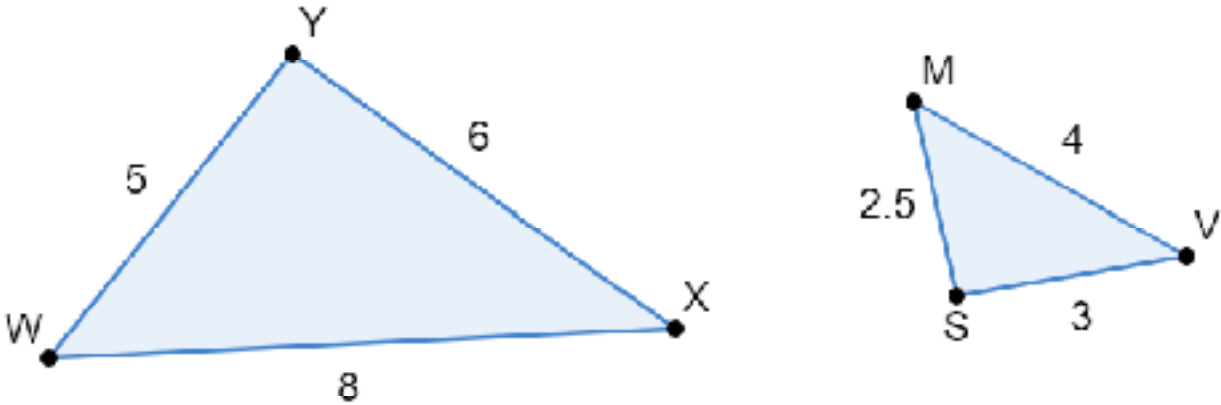
The constant of proportionality between any two similar triangles is the ratio of the corresponding side lengths. For triangle HKL and triangle MPW , the constant of proportionality is written as $\frac{\text{corresponding side length on } HKL}{\text{corresponding side length on } MPW}$.

b. Find the constant of proportionality using each set of corresponding sides from triangle *HKL* and triangle *MPW*.

corresponding side length on <i>HKL</i> corresponding side length on <i>MPW</i>	Constant of Proportionality

c. With your partner, discuss whether it is necessary to calculate the constant of proportionality for all sets of corresponding sides, or to calculate the constant of proportionality for one set of corresponding sides.

4. Triangle *WYX* and triangle *MSV* are shown.



Mansoor claims that the two triangles are not similar because the ratios of the corresponding sides are not equivalent. His work supporting his claim is shown.

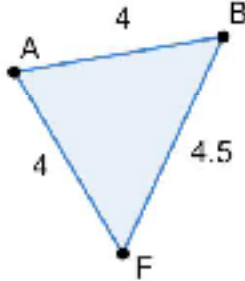
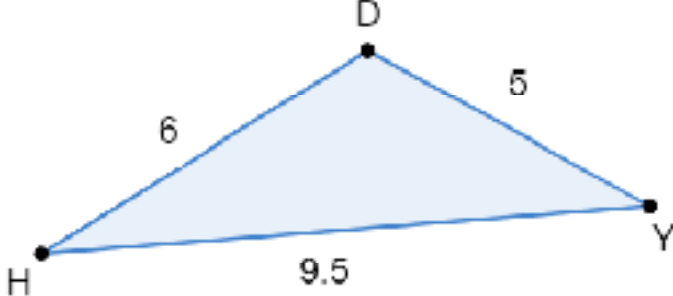
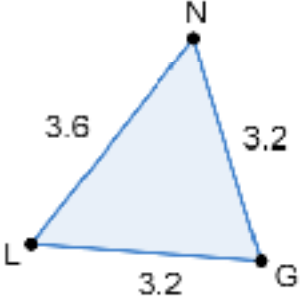
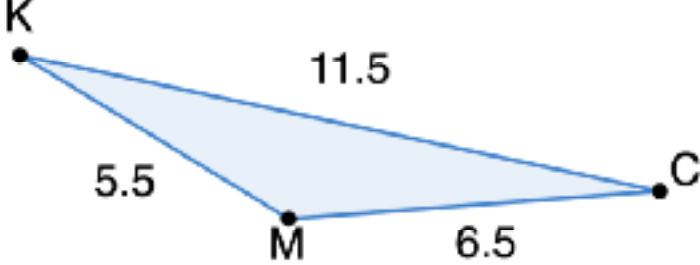
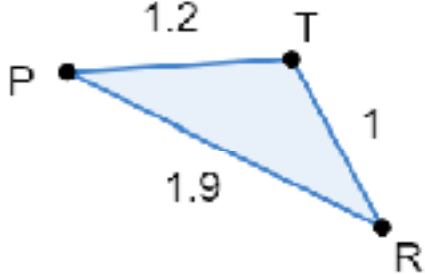
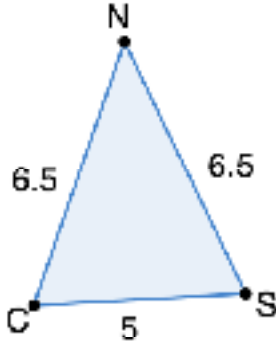
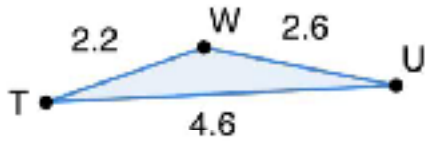
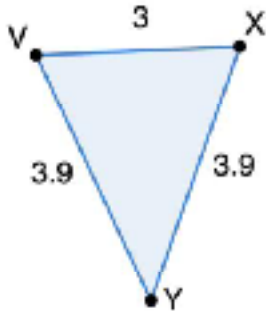
$$\frac{WY}{MS} = \frac{5}{2.5} = 2$$

$$\frac{YX}{MV} = \frac{6}{4} = 1.5$$

- a. Explain whether you agree with Mansoor. Show any work that supports your answer.
- b. Compare your work with your partner and resolve any differences, if necessary.

6.5.4 Collaboration: Matching Similar Triangles

Eight triangles are shown.

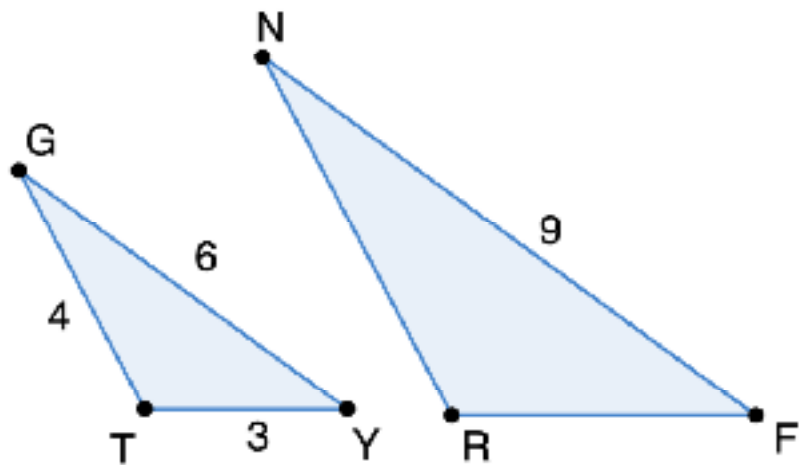
Triangle <i>ABF</i> 	Triangle <i>DHY</i> 
Triangle <i>GNL</i> 	Triangle <i>KMC</i> 
Triangle <i>PTR</i> 	Triangle <i>NCS</i> 
Triangle <i>TWU</i> 	Triangle <i>VXY</i> 

1. Work with your partner to complete the table by determining each pair of similar triangles and finding the constant of proportionality for each pair.

Similar Triangles	Constant of Proportionality
Triangles _____ and _____	
Triangles _____ and _____	
Triangles _____ and _____	
Triangles _____ and _____	

6.5.5 Guided Practice: Solving Mathematical Problems with Similar Triangles

1. Triangle *GTY* is similar to triangle *NRF*.

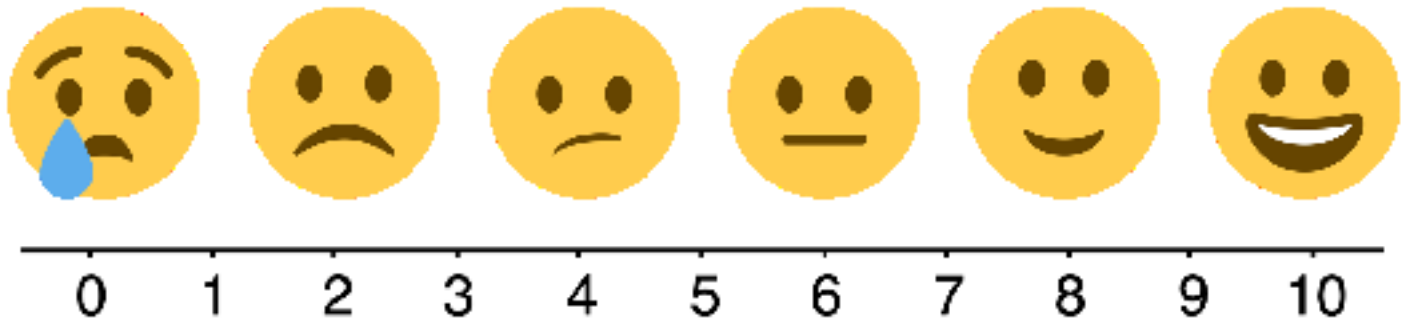


- a. What is the constant of proportionality between triangle *GTY* and triangle *NRF*?
- b. Find the lengths of the missing sides of triangle *NRF*.
- c. Complete the table to verify your work.

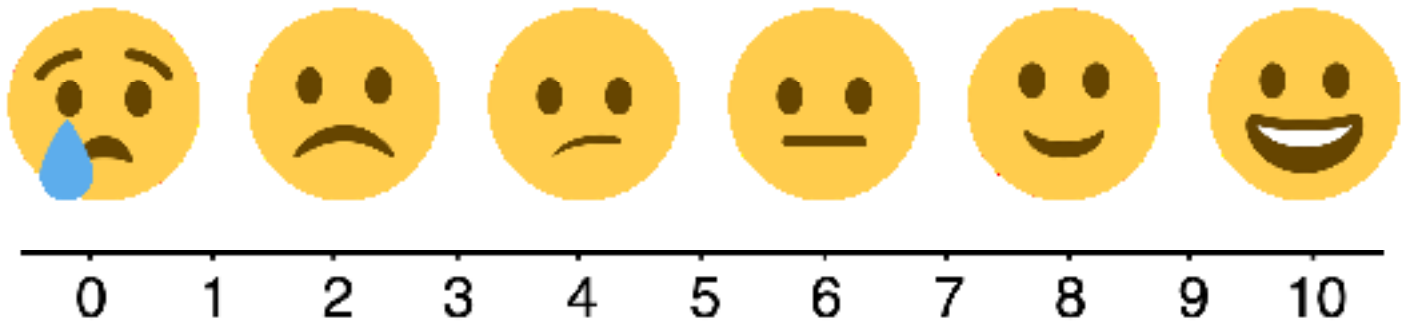
corresponding side length on <i>GTY</i> corresponding side length on <i>NRF</i>	Constant of Proportionality

6.5.6 Wrap-Up: Reflection

1. For each of the statements below, circle the emoji that represents your feelings toward each statement.
- a. I can identify the corresponding sides of similar triangles.



- b. I can determine whether two triangles are similar.



- c. I can find the missing side lengths in similar triangles.

